

Problem 6.3

At a certain rate of compound interest, \$250 deposited on July 1, 2005 has to accumulate to \$275 on January 1, 2006. Assuming the interest rate does not change and there are no subsequent deposits, find the account balance on January 1, 2008.

From July 1, 2005 to January 1, 2006 is one half-year. Therefore,

$$250(1 + i) = 275.$$

Dividing by 250,

$$1 + i = \frac{275}{250} = 1.1.$$

So the growth factor for each half-year is 1.1.

From July 1, 2005 to January 1, 2008 there are 5 half-year periods:

July 1, 2005 $\rightarrow$ Jan 1, 2006 $\rightarrow$ July 1, 2006 $\rightarrow$ Jan 1, 2007 $\rightarrow$ July 1, 2007 $\rightarrow$ Jan 1, 2008.

Thus, the balance on January 1, 2008 is

$$A = 250(1.1)^5.$$

Therefore,

$$A = 402.6275.$$

So, the account balance on January 1, 2008 is

$$\boxed{\$402.63}.$$